

LAB TASKSSS

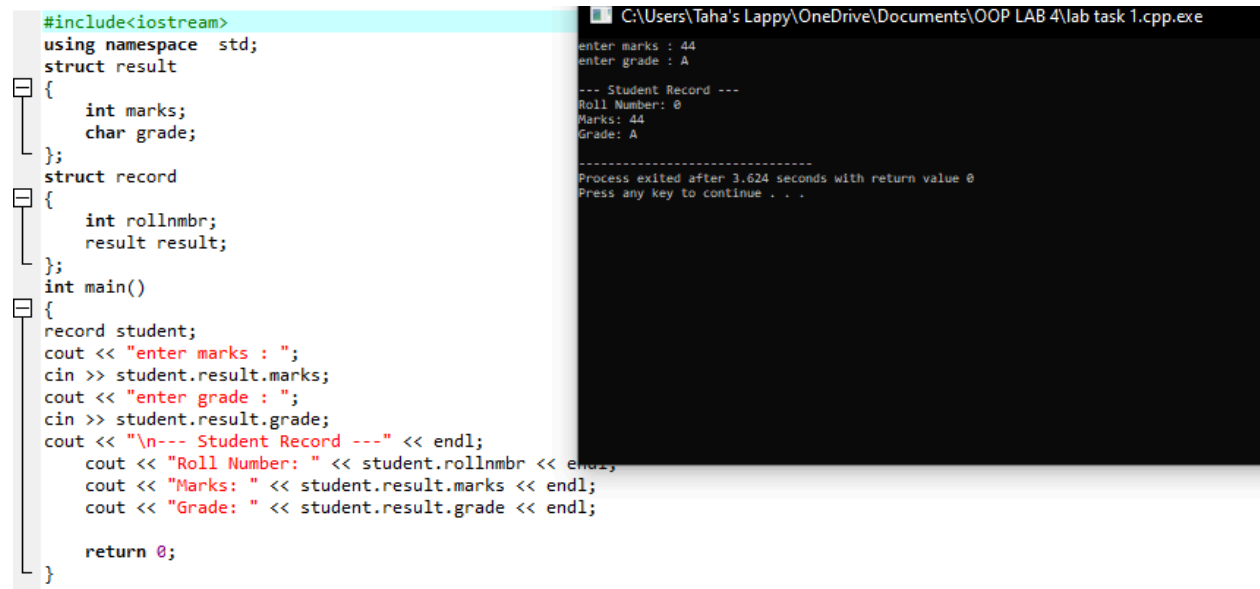
TASK 1 Write a program that uses two structures i.e., Result and Record.

The Result structure

stores marks and grade while the Record structures stores roll number and a Result type. The

program declares a variable of type Record in main, inputs roll number, marks and grade. It

finally displays these values on screen.



The image shows a C++ program in a code editor on the left and its execution output in a terminal window on the right. The program defines two structures: 'result' with 'marks' and 'grade' fields, and 'record' with 'rollnmbr' and a 'result' field. The main function prompts the user for marks and grade, then displays the student record.

```
#include<iostream>
using namespace std;
struct result
{
    int marks;
    char grade;
};
struct record
{
    int rollnmbr;
    result result;
};
int main()
{
    record student;
    cout << "enter marks : ";
    cin >> student.result.marks;
    cout << "enter grade : ";
    cin >> student.result.grade;
    cout << "\n--- Student Record ---" << endl;
    cout << "Roll Number: " << student.rollnmbr << endl;
    cout << "Marks: " << student.result.marks << endl;
    cout << "Grade: " << student.result.grade << endl;

    return 0;
}
```

Execution output:

```
C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\lab task 1.cpp.exe
enter marks : 44
enter grade : A

--- Student Record ---
Roll Number: 0
Marks: 44
Grade: A

Process exited after 3.624 seconds with return value 0
Press any key to continue . . .
```

TASK 2

Write a program that uses two structures, Data and Car. The Data structure stores

Name and Color of a car. The Car structure stores model, top speed and no. of gears in car and a

Data type. The program declares a variable of type Car, inputs the name, color, model, top speed

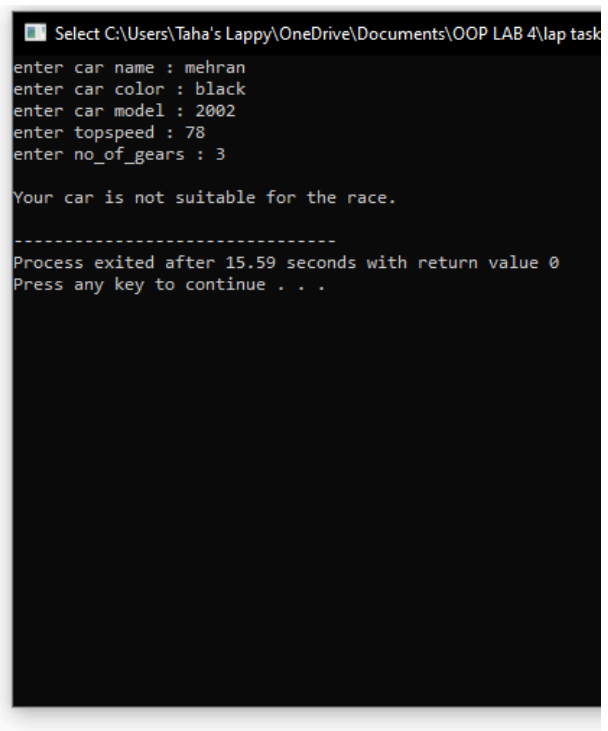
and no. of gears and display the data if top speed is greater than 200 and no. of gears are more

than 4 otherwise display that your car is not suitable for the race.

```

#include<iostream>
using namespace std;
struct data
{
    string name;
    string color;
};
struct car
{
    string model;
    int topspeed;
    int no_of_gears;
    data info;
};
int main()
{
    car c;
    cout << "enter car name : ";
    cin >> c.info.name;
    cout << "enter car color : ";
    cin >> c.info.color;
    cout << "enter car model : ";
    cin >> c.model;
    cout << "enter topspeed : ";
    cin >> c.topspeed;
    cout << "enter no_of_gears : ";
    cin >> c.no_of_gears;
    if(c.topspeed > 200 && c.no_of_gears >4)
    {
        cout << "\n    car details  ";
        cout << "Name: " << c.info.name << endl;
        cout << "Color: " << c.info.color << endl;
        cout << "Model: " << c.model << endl;
        cout << "Top Speed: " << c.topspeed << endl;
        cout << "Gears: " << c.no_of_gears << endl;
    }
    else
    {
        cout << "\nYour car is not suitable for the race." << endl;
    }
    return 0;
}

```



```

Select C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\lap task
enter car name : mehran
enter car color : black
enter car model : 2002
enter topspeed : 78
enter no_of_gears : 3

Your car is not suitable for the race.

-----
Process exited after 15.59 seconds with return value 0
Press any key to continue . . .

```

TASK 3

Declare a structure having two members with a real part and imaginary part. Program must have three functions named addition, subtract, and multiply. The program then pass the structure members to functions then add, subtract and multiply two complex numbers and then prints the result.

```

lab task 3.cpp
1 #include<iostream>
2 using namespace std;
3 struct complex
4 {
5     float real;
6     float imaginary;
7 };
8 complex addition(complex c1,complex c2)
9 {
10     complex result;
11     result.real = c1.real+ c2.real;
12     result.imaginary = c1.imaginary + c2.imaginary;
13     return result;
14 }
15 complex subtract(complex c1,complex c2)
16 {
17     complex result;
18     result.real = c1.real - c2.real;
19     result.imaginary = c1.imaginary - c2.imaginary;
20     return result;
21 }
22 complex multiplication(complex c1,complex c2)
23 {
24     complex result;
25     result.real = (c1.real * c2.real) -(c1.imaginary * c2.imaginary);
26     result.imaginary = (c1.imaginary * c2.imaginary) + (c1.imaginary * c2.imaginary);
27     return result;
28 }
29 int main()
30 {
31     complex num1,num2,sum,diff,product;
32     cout << "enter first complex number (real imaginary) : ";
33     cin >> num1.real >> num2.imaginary;
34     cout<< "enter second complex number (real imaginary) : ";
35     cin >> num1.real >> num2.imaginary;
36     sum =addition(num1,num2);
37     diff = subtract(num1,num2);
38     product = multiplication(num1,num2);
39 }

```

```

C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\lab task 3.exe
enter first complex number (real imaginary) : 33
44
enter second complex number (real imaginary) : 554
44
ADDITION : 554+444i
SUBTRACTION :554--444i
MULTIPLICATION : 7.76319e-043*0i
-----
Process exited after 6.22 seconds with return value 0
Press any key to continue . . .

```

TASK 4

Create a Library Book system using nested structures. Define a struct Author with name (string) and nationality (string). Define a struct Book with title (string), ISBN (string), price (double), publication year (int), and an Author field. In main(), declare an array of 3 Book objects. Input details for all 3 books, then display only those books published after 2015.

```

1 #include<iostream>
2 using namespace std;
3 struct author
4 {
5     string name;
6     string nationality;
7 };
8 struct book
9 {
10     string title;
11     string ISBN;
12     double price;
13     int pubyear;
14     author author;
15 };
16 int main()
17 {
18     book b[3];
19     for(int i=0;i<3;i++)
20     {
21         cout << "\nENTER DETAIL FOR BOOK: " << i+1 << endl;
22         cout << "title : ";
23         cin >> b[i].title;
24         cout << "ENTER ISBN : ";
25         cin >> b[i].ISBN;
26         cout << "ENTER PRICE : ";
27         cin >> b[i].price;
28         cout << "ENTER PUBLICATION YEAR : ";
29         cin >> b[i].pubyear;
30         cout << "ENTER AUTHOR NAME : ";
31         cin >> b[i].author.name;
32         cout << "ENTER AUTHOR NATIONALITY: \n";
33         cin >> b[i].author.nationality;
34     }
35     for(int i=0;i<3;i++)
36     {
37         if(b[i].pubyear > 2015)
38             cout << "\nTitle: " << b[i].title;
39             cout << "\nISBN: " << b[i].ISBN;
40             cout << "\nPrice: " << b[i].price;
41             cout << "\nPublication Year: " << b[i].pubyear;
42             cout << "\nAuthor: " << b[i].author.name;
43             cout << "\nNationality: " << b[i].author.nationality << endl;
44     }
45 }

```

```

C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\lab task 3.exe
europe
Title: hgghghg
ISBN: 55556
Price: 2388
Publication Year: 2020
Author: amnaa
Nationality: pakistan

Title: herry
ISBN: potter
Price: 999
Publication Year: 2111
Author: amnaaa
Nationality: india

Title: heheeee
ISBN: 99999
Price: 9876
Publication Year: 2021
Author: amnaa
Nationality: europe
-----
Process exited after 158.2 seconds with return value 0
Press any key to continue . . .

```

QUESTION 1 What is the difference between accessing a member of a simple struct vs. a nested struct? Givedot-notation examples from this task.

ANSWER

In a **simple struct**, members are accessed directly using one dot (.).

In a **nested struct**, we use **two dots** because we must go through the outer struct first.

EXAMPLE

Simple struct

b[i].title

b[i].ISBN

B[i].price

Nested struct

b[i].[author.name](#)

B[i].author.nationality

QUESTION 2

Why is it better to store the Author as a nested struct inside Book rather than having authorName and authorNationality as direct string members of Book

ANSWER

It is better because nesting groups related data, improves readability, makes the program modular, and allows the Author structure to be reused and extended easily.

QUESTION 3

If you had to add a co-author to the Book struct, how would you modify the struct definition?

Write just the updated Book struct.

ANSWER

```
struct book
{
    string title;
    string ISBN;
    double price;
    int pubyear;
    author author;    // main author
    author coAuthor;  // new co-author
};
```

TASK 5

Define a struct Student with name (string), rollNo (int), marks for 3 subjects (float array or 3

separate floats), and a gpa (float). Write a function calculateGPA() that accepts a Student by

reference and computes the GPA (average of 3 subject marks divided by 10, max 4.0).

Write a

separate displayStudent() function that accepts a Student by value and prints all details. In main(), input data for 2 students and call both functions.

```

1  #include<iostream>
2  #include<string>
3  using namespace std;
4  struct student
5  {
6      string name;
7      int rollnbr;
8      float marks[3];
9      float gpa;
10 };
11 void calculateGPA(student &s)
12 {
13     float avg = (s.marks[0] + s.marks[1] + s.marks[2]) / 3.0;
14     s.gpa = avg / 10.0;
15     if(s.gpa > 4.0)
16         s.gpa = 4.0;
17 }
18 void displayStudent(student s)
19 {
20     cout << "\n--- Student Details ---\n";
21     cout << "Name: " << s.name << endl;
22     cout << "Roll No: " << s.rollnbr << endl;
23     cout << "Marks: " << s.marks[0] << ", " << s.marks[1] << ", " << s.marks[2] << endl;
24     cout << "GPA: " << s.gpa << endl;
25 }
26 int main()
27 {
28     student students[2];
29     for(int i=0; i<2; i++)
30     {
31         cout << "\nEnter details for Student " << i+1 << endl;
32         cout << "Name: ";
33         getline(cin, students[i].name);
34         cout << "Roll No: ";
35         cin >> students[i].rollnbr;
36         cout << "Enter marks for 3 subjects";
37         cin >> students[i].marks[0];
38         cin >> students[i].marks[1];
39         cin >> students[i].marks[2];
40     }
41     for(int i=0; i<2; i++)
42         displayStudent(students[i]);
43     return 0;
44 }

```

```

C:\Users\Taha's Laptop\OneDrive\Documents\OOP LAB 4\lab ta
Enter details for Student 1
Name: amna
Roll No: 12
Enter marks for 3 subjects
33
44
55

--- Student Details ---
Name: amna
Roll No: 12
Marks: 33, 44, 55
GPA: 0

--- Student Details ---
Name:
Roll No: 4255393
Marks: 0, 5.96166e-039, 0
GPA: 9.24857e-044

Process exited after 6.88 seconds with return value 0
Press any key to continue . . .

```

Q1. What is the key difference between passing a struct by value vs. by reference? In this task, why must calculateGPA() use pass-by-reference?

Answer:

Pass-by-value: A copy of the struct is sent to the function. Changes inside the function do **not** affect the original data.

Pass-by-reference: The function receives the **actual object (memory address)**. Any changes inside the function **modify the original struct**.

Why calculateGPA() must use reference?

Because the function needs to **update the GPA inside the original Student object** in **main()**. If passed by value, GPA would change only in the copy and be lost after the function ends

.Answer:

Pass-by-value sends a copy, while pass-by-reference sends the original object. calculateGPA() must use reference so the GPA calculated inside the function updates the real Student data in main().

Q2. If displayStudent() accidentally modifies s.gpa inside the function (pass-by-value), will it affect the original struct in main()? Why or why not?

Answer: No, it will not affect the original struct because pass-by-value creates a copy, so changes occur only in the local copy inside the function.

Q3. Can a function return a struct in C++? Rewrite the signature of calculateGPA() so it returns a float instead of modifying the struct by reference.

Answer:

yes, functions can return structs or other types. The function can be rewritten to return a float GPA instead of modifying the struct by reference.

TASK 6

Create a simple ATM simulation. Define a struct Account with accountNumber (string), holderName (string), and balance (double). Write a function deposit() that takes an Account by reference and an amount, adds it to balance, and prints a receipt. Write a function withdraw() that takes an Account by reference and an amount, and returns a bool (true if successful, false if insufficient funds). Write a function createAccount() that returns a newly created Account struct filled with user input. Use all three functions in main().

```
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 struct student
5 {
6     string name;
7     int rollnmbr;
8     float marks[3];
9     float gpa;
10 };
11 void calculateGPA(student &s)
12 {
13     float avg = (s.marks[0] + s.marks[1] + s.marks[2]) / 3.0;
14     s.gpa = avg / 10.0;
15     if(s.gpa > 4.0)
16         s.gpa = 4.0;
17 }
18 void displayStudent(student s)
19 {
20     cout << "\n--- Student Details ---\n";
21     cout << "Name: " << s.name << endl;
22     cout << "Roll No: " << s.rollnmbr << endl;
23     cout << "Marks: " << s.marks[0] << ", " << s.marks[1] << ", " << s.marks[2] << endl;
24     cout << "GPA: " << s.gpa << endl;
25 }
26 int main()
27 {
28     student students[2];
29     for(int i=0; i<2; i++)
30     {
31         cout << "\nEnter details for Student " << i+1 << endl;
32         cout << "Name: ";
33         getline(cin, students[i].name);
34         cout << "Roll No: ";
35         cin >> students[i].rollnmbr;
36         cout << "Enter marks for 3 subjects";
37         cin >> students[i].marks[0];
38         cin >> students[i].marks[1];
39         cin >> students[i].marks[2];
40     }
41     for(int i=0; i<2; i++)
42         displayStudent(students[i]);
43     return 0;
44 }
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\lab ta
Enter details for Student 1
Name: amna
Roll No: 12
Enter marks for 3 subjects
33
44
55

--- Student Details ---
Name: amna
Roll No: 12
Marks: 33, 44, 55
GPA: 0

--- Student Details ---
Name:
Roll No: 4255393
Marks: 0, 5.96166e-039, 0
GPA: 9.24857e-044

Process exited after 6.88 seconds with return value 0
Press any key to continue . . .
```

Q1. What does it mean for a function to 'return a struct'? How is createAccount() in this task different

from the display/calculate functions in Lab Task 5?

Answer: Returning a struct means the function sends a complete object back to the caller. createAccount() returns a new Account, while previous functions either modified an existing struct by reference or only displayed data. _

Q2. In the withdraw() function, why is bool a better return type than void for this scenario?

Answer: bool is better because withdrawal may succeed or fail. Returning true/false allows the program to check the result and display the correct message.

Q3. If Account had a nested struct (e.g., a BranchInfo struct inside Account), how would you access the branch name inside the deposit() function? Write the access expression.

Answer: struct BranchInfo {
 string branchName;
};

```
struct Account {  
    string accountNumber;  
    string holderName;  
    double balance;  
    BranchInfo branch;  
};  
acc.branch.branchName  
acc.branch.branchName
```

HOME TASKSSSSSSSS

TASK 1

1: Write a program that uses two structures date and phonebook. The date structure stores day, month and year. The phonebook structure stores name, city, phone number and a date type. The program declares a variable of type phonebook, inputs values and then display the values.

```

using namespace std;
struct date
{
    int day,month,year;
};
struct phonebook
{
    string name;
    string city;
    string phonenumber;
    date dob;
};
int main()
{
    phonebook person;
    cout << "ENTER NAME : ";
    cin >> person.name;
    cout << "ENTER CITY : ";
    cin >> person.city;
    cout << "ENTER PHONENUMBER : ";
    cin >> person.phonenumber;
    cout << "ENTER DATE OF BIRTH : ";
    cin >> person.dob.day >> person.dob.month >> person.dob.year;
    cout << "\n ENTER PHONEBOOK RECORD : ";
    cout << "NAME : "<< person.name << endl;
    cout << "CITY : " << person.city << endl;
    cout << "PHONENUMBER : " << person.phonenumber << endl;
    cout << " DATE OF BIRTH : " << person.dob.day << "/" << person.dob.month << "/" << person.dob.year << endl;
    return 0;
}

```

```

ENTER NAME : amnaa
ENTER CITY : Lahore
ENTER PHONENUMBER : 03079879996
ENTER DATE OF BIRTH : 28feb2007

ENTER PHONEBOOK RECORD : NAME : amnaa
CITY : Lahore
PHONENUMBER : 03079879996
DATE OF BIRTH : 28/0/1

-----
Process exited after 19.5 seconds with return value 0
Press any key to continue . . .

```

HOME TASK 2

Write a program that uses three structures; Parameters, Result and Rectangle. The Parameters structure stores length and width. The Result structure stores area and perimeter and Rectangle structure stores two variables of Parameters and Results. The program declares a variable of type Rectangle, inputs length, width, calculates area and perimeter and then display The result.


```
home task 2.cpp
3 struct parameter
4 {
5     float length;
6     float width;
7 };
8 struct result
9 {
10    float area;
11    float perimeter;
12 };
13 struct rectangle
14 {
15     parameter p;
16     result r;
17 };
18 int main()
19 {
20     rectangle rect;
21     cout << "ENTER LENGTH";
22     cin >> rect.p.length;
23     cout << "Enter Width: ";
24     cin >> rect.p.width;
25
26     rect.r.area = rect.p.length * rect.p.width;
27     rect.r.perimeter = 2 * (rect.p.length + rect.p.width);
28
29     cout << "\n--- Rectangle Result ---\n";
30     cout << "Area: " << rect.r.area << endl;
31     cout << "Perimeter: " << rect.r.perimeter << endl;
32     return 0;
33 }
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\OOP LAB 4\home task 2.exe
ENTER LENGTH4
Enter Width: 4

--- Rectangle Result ---
Area: 16
Perimeter: 16

-----
Process exited after 2.791 seconds with return value 0
Press any key to continue . . .
```

HOME TASK 3